Procurement Timing & Vendor Dependence Analysis

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Problem

Organizations often analyze procurement by total spend, but not by timing, category behavior, or vendor dependence. This can obscure patterns like year-end spending concentration or reliance on a small set of suppliers and can cause an operational bottleneck called “phantom budget capacity”: where budget remaining is not operationally deployable due to approval governance and cycle time constraints.

Objective

Analyze procurement behavior across timing, category, and vendor dimensions to identify structural patterns in spending and

Data & Approach

SQL-based mart tables were created in PostgreSQL as the single source of truth. A star schema was implemented in Power BI, and weighted measures were used to ensure accurate aggregation.

Key Analyses

1. Late-Year Procurement Timing: Measured share of spend in final 30/60/90 days and identified departments with high concentration.

2. Category Timing Sensitivity: Identified categories and subcategories driving late-year purchasing.

3. Vendor Concentration: Evaluated supplier dependence and top vendor share across departments.

Key Findings

- ~23% of procurement spend occurs in the final 90 days of the fiscal year.

- Certain departments combine high spend and high late-year concentration.

- Specific categories (e.g., maintenance, institutional supplies) consistently drive year-end purchasing patterns.

- Vendor spend is highly concentrated among a small group of suppliers.

Business Implications

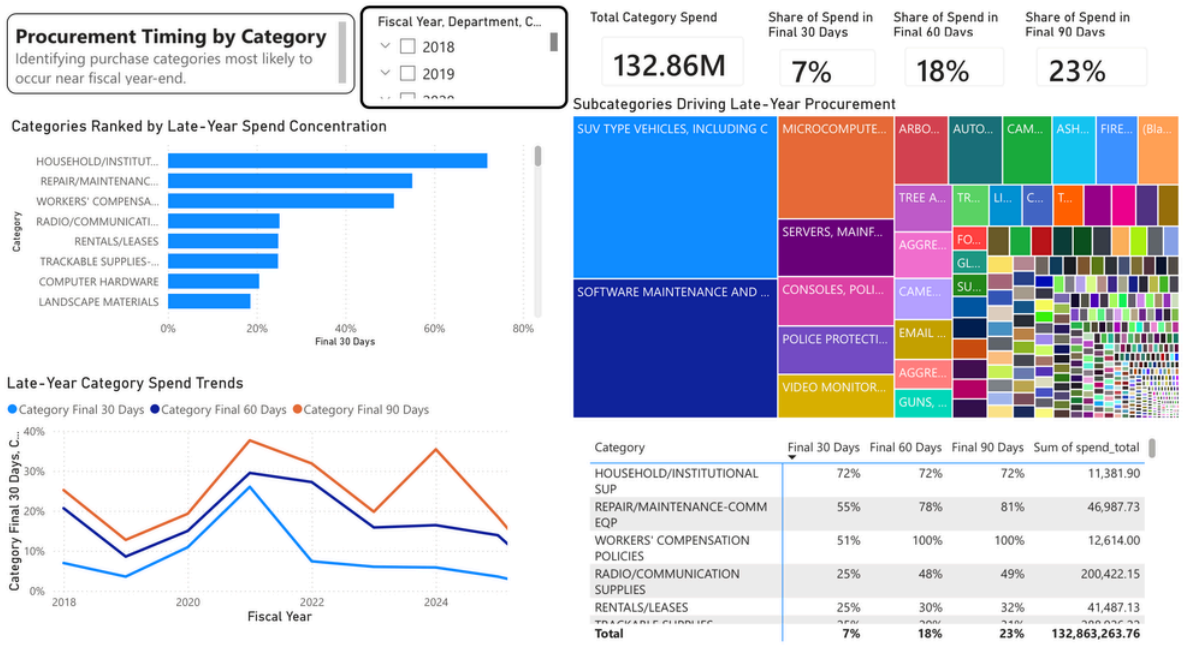
- Late-year concentration may indicate planning or budget timing inefficiencies.

- High vendor concentration suggests supplier dependency risk.

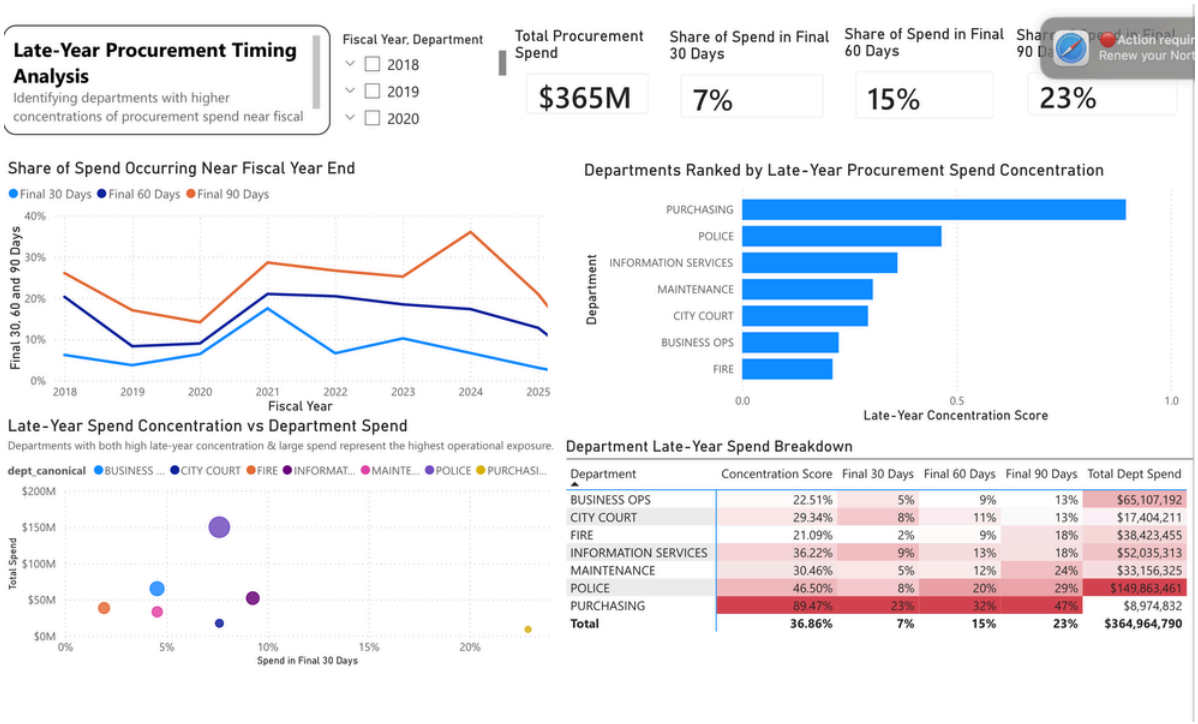
- Category patterns can inform procurement planning and contract structuring.

Tools Used PostgreSQL, SQL, Power BI, DAX

Dashboard Preview



Category	Final 30 Days	Final 60 Days	Final 90 Days	Sum of spend_total
HOUSEHOLD/INSTITUTIONAL SUP	72%	72%	72%	11,381.90
REPAIR/MAINTENANCE-COMM EQP	55%	78%	81%	46,987.73
WORKERS' COMPENSATION POLICIES	51%	100%	100%	12,614.00
RADIO/COMMUNICATION SUPPLIES	25%	48%	49%	200,422.15
RENTALS/LEASES	25%	30%	32%	41,487.13
Total	**7%**	**18%**	**23%**	**132,863,263.76**



Department Late-Year Spend Breakdown

Department	Concentration Score	Final 30 Days	Final 60 Days	Final 90 Days	Total Dept Spend
BUSINESS OPS	22.51%	5%	9%	13%	$65,107,192
CITY COURT	29.34%	8%	11%	13%	$17,404,211
FIRE	21.09%	2%	9%	18%	$38,423,455
INFORMATION SERVICES	36.22%	9%	13%	18%	$52,035,313
MAINTENANCE	30.46%	5%	12%	24%	$33,156,325
POLICE	46.50%	8%	20%	29%	$149,863,461
PURCHASING	89.47%	23%	32%	47%	$8,974,832
Total	**36.86%**	**7%**	**15%**	**23%**	**$364,964,790**

Closing

This project demonstrates how procurement data can be analyzed beyond totals — incorporating timing, composition, and dependency to better understand operational behavior.